1. Sum of array Elements

```c
#include <stdio.h>
int arr[5] = {2,4,8,1,6};
int main(){
    int sum = 0;
    int elements = sizeof(arr)/ sizeof(arr[0]);
    printf("array: { ");
    for (int i = 0; i < elements; i++){
        printf("%d ", arr[i]);
        sum += arr[i];
    }
    printf("}\nSum of array ele: %d\n", sum);
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ESP-IDF

```
PS D:\VS code\C Programming> .\sumOfArrayEle.exe
array: { 2 4 8 1 6 }
Sum of array ele: 21
PS D:\VS code\C Programming>
```

2. Big Number in Array

```c
int arr[5] = {2,4,8,1,6};
int main(){
    int bigNum = 0, elements = sizeof(arr)/ sizeof(arr[0]);
    printf("array: { ");
    for (int i = 0; i < elements; i++){
        printf("%d ", arr[i]);
        if(bigNum < arr[i]){
            bigNum = arr[i];
        }
    }
    printf("}\nBig Number In array: %d\n", bigNum);
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ESP-IDF SERIAL MONI

```
PS D:\VS code\C Programming> .\bigNum.exe
array: { 2 4 8 1 6 }
Big Number In array: 8
```

3. Pass Array into a Function

C Programming > C program.c > main()

```c
#include <stdio.h>
void printArray(int array[], int size){
    printf("Array: { ");
    for (int i = 0; i < size; i++){
        printf("%d ", array[i]);
    }
    printf("}");
}
int main(){
    int arr[5] = {2,4,8,1,6};
    int size = sizeof(arr)/ sizeof(arr[0]);
    printArray(arr, size);
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ESP-IDF

```
PS D:\VS code\C Programming> .\passArrayToFunction.exe
Array: { 2 4 8 1 6 }
PS D:\VS code\C Programming>
```

4. Insert the Element in Array at Given Pos

```
PS D:\VS code\C Programming> .\insertEleToGivenPos.exe
Enter the size of the array:
4
Enter the array elements:
array[0]:
2
array[1]:
4
array[2]:
6
array[3]:
8

Enter the Element to insert into the array:
10

Enter the pos where element to insert from 1 to 4
4
2 4 6 10 8
PS D:\VS code\C Programming>
```

5. Merging two array Into a single array

```
PS D:\VS code\C Programming> .\merge.exe
Enter the len of array1:
3
Enter the len of array2:
2
Enter the array 1 Elements
Array1[0]
1
Array1[1]
3
Array1[2]
5
Enter the array 2 Elements
Array2[0]
7
Array2[1]
9
Merging the array

Array 1:
1 3 5
Array 2:
7 9
Merged array:
1 3 5 7 9
PS D:\VS code\C Programming>
```

6. Merging two arrays into a single array

```
PS D:\VS code\C Programming> .\linearSearch.exe
{ 10 20 30 30 50 60 70 80 90 100  }
Enter Element to find in the array: 25
Element not found
PS D:\VS code\C Programming> .\linearSearch.exe
{ 10 20 30 30 50 60 70 80 90 100  }
Enter Element to find in the array: 30

Entered Element 30 Found at 3
PS D:\VS code\C Programming>
```

7. Bubble Sort

```c
void BubbleSort(int array[], int len){
    for (int i = 0; i < len; i++){
        for (int j = 0; j < (len-1) ; j++){
            if (array[j] > array[j+1]){
                int temp = array[j];
                array[j]  =array[j+1];
                array[j+1] = temp;
            }
        }
        len -=1;  // which reduce the compile
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ESP

```
PS D:\VS code\C Programming> .\bubbleSort.exe
Given array is:
1 4 2 3 6 5 7 9 8
Sorted array is:
1 2 3 4 5 6 7 8 9
PS D:\VS code\C Programming>
```

8. Selection Sort

C Logical programming > C program.c > selectionSort(int [], int)

```c
void selectionSort(int arr[], int n) {
    int i, j, minIndex, temp;

    for (i = 0; i < n - 1; i++) {
        minIndex = i; // Assume current index is the minimum

        // Find the index of the smallest element in the remaining
        for (j = i + 1; j < n; j++) {
            if (arr[j] < arr[minIndex]) {
                minIndex = j;
            }
        }

        // Swap the found minimum element with the first element
        if (minIndex != i) {
            temp = arr[i];
            arr[i] = arr[minIndex];
            arr[minIndex] = temp;
        }
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Social Media Automation\C Logical programming> .\selectionSort.exe
Original array: 64 25 12 22 11
Sorted array:   11 12 22 25 64
PS D:\Social Media Automation\C Logical programming>
```

9. Insertion Sort

```c
void Insertion_Sort(int arr[], int len) {
    int i, j, temp;
    for (i = 1; i < len; i++) {
        temp = arr[i];  // current element to insert
        j = i - 1;

        // shift larger elements to the right
        while (j >= 0 && arr[j] > temp) {
            arr[j + 1] = arr[j];
            j--;
        }

        // place temp in correct position
        arr[j + 1] = temp;
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Social Media Automation\C Logical programming> .\insertionSort.exe
Array before Insertion sorting is:
12 10 23 25 22 50 34
Sorted array is:
10 12 22 23 25 34 50
PS D:\Social Media Automation\C Logical programming>
```